

Portfolio

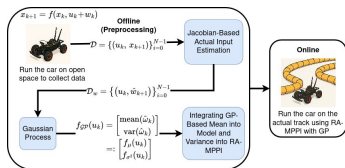
Hanif Edma

Research and engineering projects

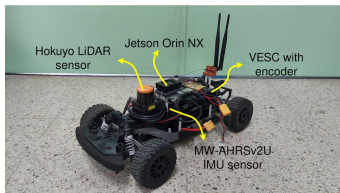
See it live: <https://hanifedma.com/portfolio>

Performance-Enhanced Risk-Aware MPPI using Gaussian Process

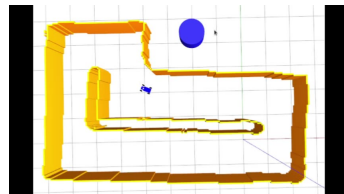
RESEARCH



Method framework



F1TENTH platform



Control demo

Performance-Enhanced Risk-Aware MPPI using Gaussian Process

RESEARCH

Published

1st author

IEEE Access

SCI Q1

2025

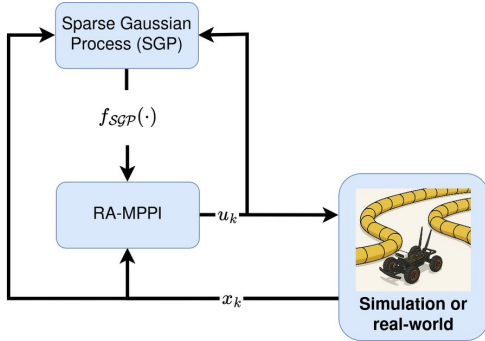
- Trains a Gaussian Process to learn the gap between a robot's nominal model and its true dynamics.
- Captures both the mean deviation and its uncertainty, feeding them into a risk-aware MPPI controller.
- Improves tracking accuracy and safety over baseline controllers.
- Validated in simulation and on real hardware (F1TENTH and Clearpath Jackal).

Images, video & explanation: <https://gp-rampipi.github.io/>

Paper (IEEE Access): <https://doi.org/10.1109/ACCESS.2025.3640166>

Robust RA-MPPI using Online Learning

RESEARCH



Control scheme



Real-world demo

Robust RA-MPPI using Online Learning

RESEARCH

Submitted to IJCAS (Springer)

1st author

- Combines risk-aware MPPI control with online learning.
- A Sparse Gaussian Process identifies and quantifies model–reality discrepancies during operation.
- The controller uses this learned uncertainty to improve robustness and collision avoidance.
- Demonstrated on an autonomous vehicle with better trajectory tracking than standard MPPI.

Images, video & explanation: <https://sgp-mppi.github.io/>

ESP32 IoT Medical Gateway

ENGINEERING PROJECTS



ESP32 IoT Medical Gateway

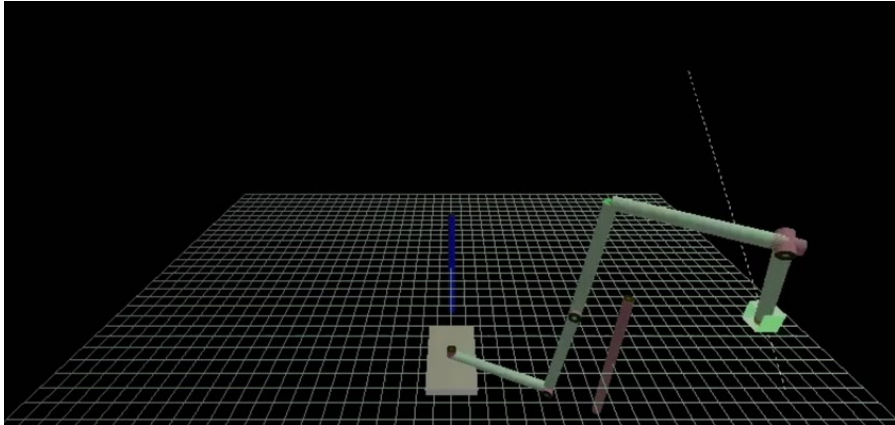
ENGINEERING PROJECTS

Internship Team of 2 Embedded / IoT ESP32 2022

- Built as an IoT Engineer Intern at DoctorTool (PT Medifa Infoyasa Suryantara).
- Developed an ESP32-based IoT gateway bridging medical instruments to the cloud.
- Connects thermometers, oximeters, and biometric sensors.
- Supports over six device categories.
- Built as a team of 2 people.

6-DOF Robot Manipulator Simulation (OpenGL)

ENGINEERING PROJECTS



6-DOF Robot Manipulator Simulation (OpenGL)

ENGINEERING PROJECTS

Robotics course project

Robot kinematics

C / OpenGL

Universitas Indonesia

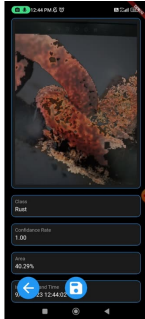
May 2023

- Built as a project for a robotics course at Universitas Indonesia.
- Real-time 3D visualization of a 6-DOF robot arm, built in C with OpenGL (GLUT).
- Implements forward kinematics and Jacobian pseudoinverse-based inverse kinematics for end-effector position control.
- Generates linear point-to-point trajectories between commanded poses.
- Streams joint commands to the physical robot over serial communication.

Source code: <https://github.com/hanifedma/robot-6dof-opengl>

Corrosion Detection App (Computer Vision)

ENGINEERING PROJECTS



In-app rust detection



1st place at SciHackfest 2023

Corrosion Detection App (Computer Vision)

ENGINEERING PROJECTS

1st place

Hackathon

Prize 25M IDR (~\$1,600)

Team of 3

Flutter

Computer Vision

2023

- Built a computer-vision system for corrosion detection.
- Shipped the Android app with Flutter (also runs on Windows and Linux).
- Delivered within a one-month timeframe.
- Won 1st place at SciHackfest 2023 (a hackathon held by PT Sucofindo) — prize 25 million IDR (~\$1,600), as a team of 3.